

Construction of the Composite Governance Index

Study: Awareness and training offerings regarding digitalisation and artificial intelligence (AI) within German healthcare provision

This document records which questionnaire items the governance index is built from, how each component was coded, how the composite was formed, and the reasoning behind these decisions. It is intended to make the index reproducible from the cleaned data set.

1. Conceptual basis

The index follows the Shadow IT governance literature, in which effective governance is treated as integrative: it combines the resolution of system gaps through sanctioned alternatives, general policy and awareness measures, and instance-specific oversight, rather than relying on any single measure in isolation (Klotz et al., 2019). The index is therefore designed to capture the number of distinct, employer-provided governance instruments a physician works under, rather than the strength of any one of them.

Klotz et al. (2019) distinguish four general governance dimensions. Three of these correspond to instruments a physician can observe and report: the provision of sanctioned AI tools (systems gap resolution), AI-specific training (awareness training), and written guidelines on AI use (policy setup). The fourth dimension, monitoring and identification (instance-specific oversight), is omitted, because whether and how an employer monitors or audits AI use is not reliably observable through physician self-report; including it would introduce unreliable measurement rather than information.

2. Component items and their sources

Each retained dimension is operationalised by a single questionnaire item. The source columns refer to the raw Microsoft Forms export.

Governance dimension (Klotz et al., 2019)	Employer-provided instrument	Questionnaire item	Source column
Systems gap resolution	Provision of AI tools	Q14: Does your employer provide you with AI-based applications?	AL
Awareness training	AI-specific training	Q22: Have you received training specifically on AI in a professional context?	AT
Policy setup	Written guidelines on AI use	Q33: Have you received a guideline or similar from your employer on the use of AI, in particular LLMs?	BE
Instance-specific oversight (monitoring and identification)	Omitted	Not measured	—

3. Coding of the components

Each component is a binary indicator: 1 where the measure is in place, 0 otherwise. Responses of 'do not know' or 'cannot say' were coded as the measure not being in place. The reasoning is behavioural: a governance measure that is unknown to the physician cannot shape that physician's behaviour and is, for the purpose of explaining individual conduct, functionally absent. This is a deliberately conservative rule that aligns the index with the role it plays in the analysis, namely accounting for behaviour rather than recording the formal existence of a measure.

For written guidelines (Column BE), free-text responses were first recoded to Yes, No or Don't know according to the rules set out in the data cleaning documentation before the binary coding below was applied.

Component item	Response option	Code
Provision of AI tools (Q14)	Yes	1
	No	0
	I don't know / cannot say	0
AI-specific training (Q22)	Yes	1
	No	0
	No answer	0
Written guidelines (Q33)	Yes	1
	No	0
	I don't know	0
	Other (non-substantive, after free-text recoding)	0

4. Formation of the composite index

The composite index is the unweighted sum of the three binary components and ranges from 0 (no measure in place) to 3 (all three in place).

The components are weighted equally for three reasons: there is no empirical basis in the literature for differential weighting; the three instruments are conceptually distinct and non-substitutable, so none should dominate by construction; and an unweighted count is transparent and directly interpretable as the number of measures a physician works under.

The index is a formative composite: the three measures jointly constitute the construct (the extent of employer-provided AI governance) rather than reflecting an underlying latent trait. Consequently, reflective-scale procedures such as internal-consistency reliability (for example Cronbach's alpha) and factor analysis are not appropriate, and none is reported. The relevant form of scrutiny is coverage of the governance dimensions, addressed in Sections 1 and 2.

4.1 Index used descriptively; components used as predictors

The composite index (0 to 3) is used descriptively, to summarise the distribution of governance provision across healthcare settings. The regression models do not use the composite; they enter the three binary components as separate predictors. This separation is intentional: it

identifies which individual instrument is associated with Shadow AI exposure, which a single summed score would obscure.

5. Realised values in the analytical sample (n = 320)

Prevalence of each component (coded 1):

Component	In place (% of n = 320)
Provision of AI tools (Q14)	30.63
AI-specific training (Q22)	18.44
Written guidelines (Q33)	8.75

Distribution of the composite index (overall):

Index value	Share of sample (%)	Interpretation
0	60.31	No measure in place
1	26.25	One measure in place
2	8.75	Two measures in place
3	4.69	All three measures in place

